Relation between the Boardroom Centrality and the Firm Performance

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ABSTRACT

How managers' social network would affect a firm's performance, and what kind of relationship between them are two main points we discuss during our research. We evaluate a manager in four centralities: degree centrality, closeness centrality, betweenness centrality, and eigenvector centrality, and compare to their company performance. Overall, our results suggest that executives' social networks relate tightly to firm performance and have a positive relation to it.

CCS CONCEPTS

• Applied computing; **• Enterprise computing**; **• Business process management**; **• Business intelligence**;

KEYWORDS

Additional Keywords and Phrases:Social Network, Network Centralities, Firm Performance

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1 INTRODUCTION

Social network plays an important role in many aspects of our life especially in career field. A company's performance is largely determined by the managers who runs the company. Thus, the manager's ability and social networking have a strong influence on the company. Research on the relation between boardroom centrality and the firm's performance is worth to spend time doing so. We evaluate the manager's social network in centrality from four dimensions. They are degree centrality, closeness centrality, betweenness centrality, and eigenvector centrality, which depend on manager's study experience, hometown city, company stock code, and positions they are in the company. We assume that, for example, if two persons graduated from the same university, then they probably know each other. We analyze data in both individual level and firm level to see the significance of the variables. Also, we analyze individual and firm-level together to see more about the relation between individual network and firm performance. We evaluate firm's performance into five variables, firms' ROA, ROE, SIZE, LEV, and REV_G. We find the correlation of them, and make the T-test, find out most of the variables are significant to firm's performance, and we can conclude that, manager's social network is a positive proportion to firm's performance. Overall, we observe the better the manager's social network, the better the firm's performance.

2 LITERATURE

Delving into the effect of social network among company executives and company's future is always an interesting topic to study. There is a huge number of ones about this topic.

Larcker et al. (2013) studied the relations between a board's well-connectedness and the firm's future performance in the article "Boardroom centrality and firm performance", and suggested that director networks provide economic benefits that are not immediately reflected in stock prices. They used five measures of a board's well-connectedness, and found that firms with the best-connected boards on average earn substantially higher future excess returns compared to firms with worst-connected boards. [1] El-Khatib et al. (2015) suggested that "corporate decisions can be influenced by a CEO's position in the social hierarchy, which high-centrality CEOs using their power and influence to increase entrenchment and reap private benefits". [2] Engelberg et al.(2012) found out act social networks lead to either better information flow or better monitoring. [3] Chemmanur et al.(2019) conclude that "firms with higher quality top management teams are able to develop a larger number of both exploratory and exploitative innovations.". [4] Helmers et al.(2017) "examine the effect of board interlocks on patenting and R&D spending for publicly traded companies in India", and they find that "board interlocks have significant positive effects on both R&D and patenting". [5] Ahern (2017) in the research showing that inside traders earn prodigious returns of 35% over 21 days, with more central traders earning greater returns". [6] Wong et al.(2015) present evidence that "board interlocks are positively linked with similarities in the design of executive compensation packages in interlocked firms, particularly the proportions of the options component".[7] Hwang & Kim (2009) suggests that "social ties do matter and consequently, that a considerable percentage of the conventionally independent boards are substantively not". [8] Cai et al., (2016) find "a significant positive relation between stock transaction costs and a company's social ties to the investment community". [9] Bajo et al. (2016) use a hand-collected data set of pre-IPO media coverage as a proxy for investor attention, and show that an important channel thought which more central lead IPO underwriters achieve favorable IPO characteristic is by attracting greater investor attention the IPOs underwritten by them. [10]

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3 NETWORK MODELING

To analysis the relations between executives, we built a model by constructing a matrix to find out if the executives have the same parameters. We considered four factors to see whether two of the executives are related or know each other. The four factors are past working experience, education experience, related listed companies, and geographic location. Past working experience is the companies a person used to work; education experience refers to the universities an executive attended; related listed companies are the places that an executive related; geographic location is to see whether two executives are from the same city, which might know each other. We put four parameters together, (exp, edu, stkcd, loc), and represented by 0, not related, and 1, related. For example, if two people know each other because they were graduated from the same university, then it will show like (0,1,0,0). If four of these are all 0, then they probably don't know each other.

After processing those data, we need to analyze them by measuring their centralities, which represented to their importance in the network. Using data to directly show who is the most significant person in the network.

Degree centrality measures the number of persons an executive knows in social network. The way to measure the degree centrality is by calculating the number of vertices which connected to other vertices. Mathematically, it represented as the total vertices subtract 1, n-1. This is a simple and efficacious way to directly measures the quantity of person an executive knows.

Closeness centrality measures the distance that an executive to others in social network. The mean to calculate the closeness centrality is 1 divide the sum of all the distance the person to others. The maximum number it theoretically can be is 1/(n-1). The real-life effect of this measurement represents the more important the person, the shorter the distance to others. Also, closeness importance can be understood as the time spent for sending information from a vertex to the other.

Betweenness centrality measures the times that a vertex plays a role as a bridge in the shortest route. For every vertex, the way to calculate the betweenness can be done in the following steps. First, for each pair of vertices (s, t), calculate the total shortest route, then calculate the proportion of the routes that pass through the vertices v. Last, sum them up. For directed graph, the maximum result could be (n-1) (n-1), while for undirected graph, the max result could be (n-1) (n-2)/2.

Eigenvector centrality represents that the importance of vertices not only depend on the degree centrality, but depend on the adjacent vertices' importance. We may use the adjacency matrix to find eigenvector centrality. In general, there will be many different eigenvalues for which a non-zero eigenvector solution exists. Since the entries in the adjacency matrix are non-negative, there is a unique largest eigenvalue, which is real and positive. This greatest eigenvalue results in the desired centrality measure.

4 SUMMARY STATISTICS

4.1 A Motivating Example

After we process the data, and the result shows us a networking of the executive in companies. We select two examples of the most important people in the group. The person whose user Id is 27596, Faping Zhong, was worked in four universities, which are Central South University, Hunan University, Tsinghua University and University of Chinese Academy of Sciences. He works in Physics department. He is the Chairman of a company which stock code is 600478. The other example is an executive whose user Id is 13096, graduating from China Europe International Business School, whose name is Mingzhu Dong. The universities she attended are Nanjing Tech University, Nanjing University of Science and Technology, and University of Science and Technology of China. Dong is the chairman of the company, and her birthplace is Nanjing, Jiangsu, which is a big city.

4.2 Graph

We use all the data we have collected to generate the graph of networking of the management. As shown in Figure 2, with a group of people cluster in the center who know each other, while many data remain at peripheral who are solitary. We choose the center data to make a deeper analysis. Only about 2% of the data are in the center, what makes this happen is because we only use the data in 2016, so that most of the managements have no relation with others.

We clear all the vertices with no connections with others. However, we found the sizes of the vertices are the same. We make some adjustments to the vertices. The size of the vertices is proportional to the degree of centrality. The larger the degree of centrality, the bigger the vertices.

The width of the nodes is related to the betweenness centrality. Amount the four parameters, only the betweenness centrality has the property of the nodes. The wider the edge, the more the betweenness centrality. Lastly, we generated the final graph of the network of the top management team which can manifest clearly of the network. As shown in Figure 1.

4.3 Summary Statistics

4.3.1 *Variable Definitions.*

4.3.2 Individual Level Summary. In Table 1, we analyze 10126 sample's four centrality variables. The mean of DEGREE is 0.000174, and the standard deviation of the distribution is 0.000748. The mean of DEGREE indicates there are approximately 1.76 persons an individual knows. We admit the number of 1.76 is lower than expected. The reasons are, firstly, we only consider the sample from year 2016 in our analysis and we do not include profile information prior to that year; secondly, there might be other channels that top manages may know each other besides discussed education and professional experience connections. The mean of CLOSENESS is 0.048430 with a standard deviation of 0.172319. That means, on average, the distance between two managers in our graph is about 20, indicating two random persons could be connected by 20 steps of connection. This is slightly lower than our expectation given our sample selection limitation. The median of CLOSENESS is 0 in our results. The mean of BETWEENNESS is 0.000002 with a standard deviation of 0.000034, which refers to in average, one manager plays a role for a 'bridge' for 0.020252 times theoretically. The mean of Eigenvector is 0.000832 with the standard deviation of 0.009903.

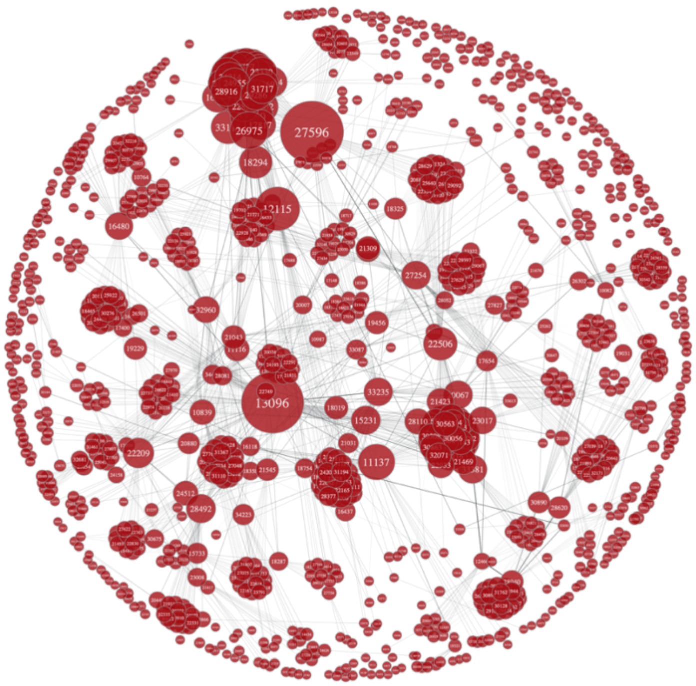

Figure 1: Network Graph of Top Management Team

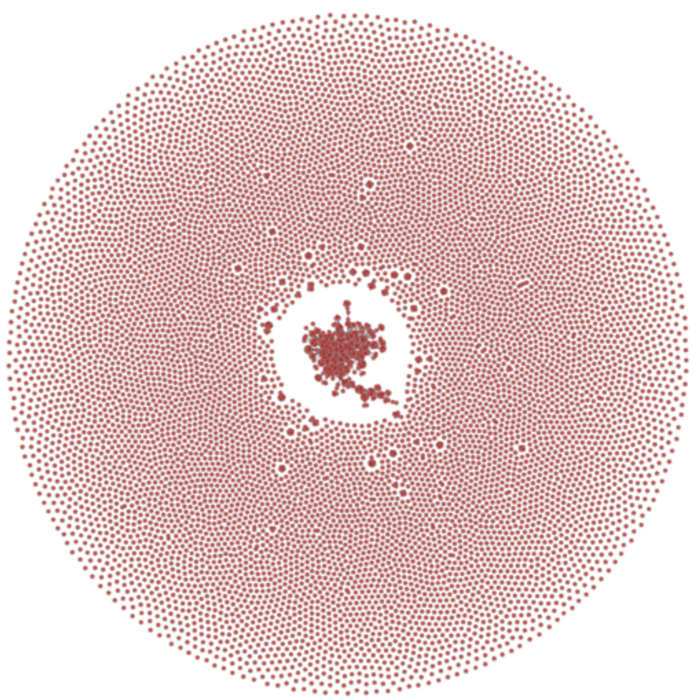

Figure 2: Network Graph of All Management Team

Variable	Definition
DEGREE	Degree centrality for manager i, standardized to (0,1)
DEGREE_M	Degree centrality for firm, j, measured as the average of all managers' degree centralities in firm j
CLOSENESS	Closeness centrality for manager i, standardized to (0,1)
CLOSENESS_M	Closeness centrality for firm, j, measured as the average of all managers' closeness centralities in firm j
BETWEENNESS	Betweenness centrality for manager i, standardized to (0,1)
BETWEENNESS_M	Betweenness centrality for firm, j, measured as the average of all managers' betweenness centralities in firm j
EIGENVECTOR	Eigenvector centrality for manager i, standardized to (0,1)
EIGENVECTOR_M	Eigenvector centrality for firm, j, measured as the average of all managers' eigenvector centralities in firm j
ROA	Return on asset, which is calculated as the division between firm's net income and total asset.
ROE	Return of equity, which is calculated as the division between firms' net income and total equity.
LEV	Firm leverage ratio, which is calculated as firm total liability divided by total asset.
SIZE	Firm size, which is calculated as the natural log of total asset.
REV_G	Firm revenue growth, which is calculated as the change of revenue of t scaled by revenue at t-1.

Table 1: Statistics of Centrality Variables

Variable	N	Mean	SD	P1	P25	Median	P75	P99
DEGREE	10126	0.000174	0.000748	0.000000	0.000000	0.000000	0.000000	0.004543
CLOSENESS	10126	0.048430	0.172319	0.000000	0.000000	0.000000	0.000000	1.000000
BETWEENNESS	10126	0.000002	0.000034	0.000000	0.000000	0.000000	0.000000	0.000033
EIGENVECTOR	10126	0.000832	0.009903	0.000000	0.000000	0.000000	0.000000	0.003973

Table 1 reports the summary statistics of four centrality variables at individual level. DEGREE is measuring the degree centrality. CLOSENESS is measuring the closeness centrality. BETWEENNESS is measuring the betweenness centrality. EIGENVECTOR is measuring the eigenvector centrality.

Table 2: Statistics of Centrality Variables between firms

Variable	N	Mean	SD	P1	P25	Median	P75	P99
DEGREE_M	3118	0.000173	0.000463	0.000000	0.000000	0.000000	0.000049	0.002272
CLOSENESS_M	3118	0.049143	0.113348	0.000000	0.000000	0.000000	0.054082	0.500000
BETWEENNESS_M	3118	0.000002	0.000023	0.000000	0.000000	0.000000	0.000000	0.000063
EIGENVECTOR_M	3118	0.000785	0.005706	0.000000	0.000000	0.000000	0.000000	0.042234

Table 2 reports the summary statistics of four centrality variables at firm level. DEGREE_M is measuring the degree centrality. CLOSENESS_M is measuring the closeness centrality. BETWEENNESS_M is measuring the betweenness centrality. EIGENVECTOR_M is measuring the eigenvector centrality.

4.3.3 Firm Level. In Table 2, we analyze 3118 firms' four centrality variables. The mean of DEGREE_M is 0.000173, and the standard deviation of the distribution is 0.000463. The mean of CLOSENESS_M is 0.049143 with the standard deviation of 0.113348. The mean of BETWEENNESS_M is 0.000002 with the standard deviation of 0.000023. The mean of EIGENVECTOR_M is 0.000785 with the standard deviation of 0.005706.

4.3.4 Firm-Year-Quarter Level. After merging centrality data with firm fundamental data, we finally have 153,810 observations for our empirical analysis. The firm performance-related observations are smaller than expected and the reasons are two-folds. 1) Some firms may have missing fundamental data; 2) Some observations are lost due to lagging when calculate the revenue growth indicator. Our final sample starts at year 2000 and ends at year 2019. The number of unique firms in our sample is 3118. The average ROA is 0.0238, which indicate the firm return on asset ratio is 2.38% on average; the average ROE is 4.29%; the average leverage ratio is around 45.99%; and the average of revenue growth is 24.61%.

4.3.5 Correlation Matrix. From Table 4, the correlation between Degree centrality and ROA is 0.0166, which means the ROA is positively related to degree centrality. The correlations between Degree centrality and ROE, SIZE, REV_G, are 0.025, 0.0837, and 0.0072, respectively, suggesting our degree centrality measure is positively correlated with main firm performance measure. Betweenness centrality and revenue growth do not have strong correlation and the correlation coefficient is near 0.00.

Table 3: Summary Statistics at Firm-Year-Quarter Level

Variable	N	Mean	SD	P1	P25	Median	P75	P99
ROA	153548	0.023801	0.040990	-0.137572	0.004736	0.017219	0.040018	0.169594
ROE	153483	0.042893	0.090061	-0.417999	0.010673	0.034914	0.074778	0.356582
LEV	153553	0.459996	0.226994	0.046743	0.287133	0.455022	0.616355	1.156884
SIZE	153558	21.822169	1.409907	19.011974	20.842758	21.635056	22.562878	26.704235
REV_G	140896	0.246125	0.759499	-0.784156	-0.042851	0.117008	0.318528	5.603276
DEGREE_M	153810	0.000178	0.000472	0.000000	0.000000	0.000000	0.000049	0.002272
CLOSENESS_M	153810	0.048953	0.110584	0.000000	0.000000	0.000000	0.055453	0.500000
BETWEENNESS_M	153810	0.000003	0.000026	0.000000	0.000000	0.000000	0.000000	0.000077
EIGENVECTOR_M	153810	0.000817	0.005706	0.000000	0.000000	0.000000	0.000000	0.042234

Table 3 reports the summary statistics of the firm performance and firm's centrality variables. ROA refers to Return on Assets. ROE refers to Return on Equity. These two variables measure the effectiveness of the jobs done by the company's management team. LEV refers to leverage ratio. DEGREE_M is measuring the degree centrality. CLOSENESS_M is measuring the closeness centrality. BETWEENNESS_M is measuring the betweenness centrality. EIGENVECTOR_M is measuring the eigenvector centrality.

Table 4: Correlation Matrix of variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) ROA	1							
(2) ROE	0.7136	1						
(3) LEV	-0.3550	-0.0795	1					
(4) SIZE	0.0569	0.1268	0.3222	1				
(5) REV_G	0.1281	0.1283	0.0411	0.029	1			
(6) DEGREE_M	0.0166	0.0250	0.0046	0.0837	0.0072	1		
(7) CLOSENESS_M	0.0451	0.0300	-0.043	0.0289	0.0050	0.3798	1	
(8) BETWEENNESS_M	0.0111	0.0296	0.0363	0.0621	-0.0000	0.4496	0.1049	1
(9) EIGENVECTO_M	0.0027	0.0062	-0.0033	0.0524	0.0052	0.6415	0.1444	0.0767

Table 4 reports the correlation between variables of firm's performance and centrality variables. ROA refers to Return on Assets. ROE refers to Return on Equity. DEGREE_M is measuring the degree centrality. CLOSENESS_M is measuring the closeness centrality. BETWEENNESS_M is measuring the betweenness centrality. EIGENVECTOR_M is measuring the eigenvector centrality.

5 UNIVARIATE ANALYSIS

5.1 T-test

From the results above in Table 5. The variables in rows are Centrality variables, which are degree centrality, closeness centrality, betweenness centrality and eigenvector centrality. In the variables in columns are Firm performance, which are return of assets, return of equity, LEV, size, REV_G. We divided data into 20 groups with for row variables and five column variables. To implement the T-test, we consider the variables in two ways, which are High group with Centrality greater than 0, and Low group with Centrality equal to 0.

Horizontally, we think of one centrality variable under different firm performance indicators. For example, in the first row in Table5, when we think of degree centrality under different firm performances indicators, we can conclude that under the degree centrality, ROA in High group is greater than that in Low group. The difference is 0.002, which means that the average ROA in high group is roughly 0.2% higher than that in low group. And the difference is statistically significant at 1% level (T-stat=-12.02). Similar to ROE, SIZE and REV_G, their degrees of centrality are all greater in High group than that in Low group. ROE in High group is greater than that in Low group. The difference is 0.001, which means that the average ROE in high group is roughly 0.1% higher than that in low group. And the difference is statistically significant at 1% level (T-stat=-11.12). Yet, there is slightly different result in LEV. LEV in High group is less than that in Low group. The difference is -0.01, which means that the average LEV in high group is roughly 1% lower than that in low group, and the difference is statistically significant at 1% level (T-stat=5.56)Now only with the degree centrality variable, the remaining three variables, Closeness centrality, betweenness centrality and eigenvector centrality are also have the same pattern.

Vertically, we think of one firm performance indicator under different centralities. For example, from the first column in Table 5, we can see firm's return of assets in different centrality variables. The absolute value of T-stat is smaller in eigenvector centrality and betweenness centrality than that in degree centrality and closeness centrality. Whatever way we looked into the table, the results and conclusion come out is similar. The statistical significance under the same firm performance indicator are similar under different centralities.

Table 5: By group T-stats table

		(1)	(2)	(3)	(4)	(5)
		ROA	**ROE**	**LEV**	**SIZE**	**REV_G**
DEGREE	Low = 0	0.0230	0.0413	0.4621	21.7690	0.2408
	High = 1	0.0257	0.0469	0.4548	21.9564	0.2595
	Diff	0.00***	0.01***	-0.01***	0.19***	0.02***
	T-stat	-12.02	-11.12	5.65	-23.52	-4.17
CLOSENESS	Low = 0	0.0230	0.0413	0.4621	21.7690	0.2408
	High = 1	0.0257	0.0469	0.4548	21.9564	0.2595
	Diff	-0.00***	-0.01***	0.01***	-0.19***	-0.02***
	T-stat	-12.03	-11.12	5.65	-23.52	-4.17
BETWEENNESS	Low = 0	0.0236	0.0425	0.4593	21.8043	0.2454
	High = 1	0.0273	0.0539	0.4751	22.1884	0.2615
	Diff	0.00***	-0.01***	-0.02***	-0.38***	-0.02*
	T-stat	-7.47	-10.56	-5.77	-22.52	-1.68
EIGENVECTOR	Low = 0	0.0235	0.0419	0.4595	21.7684	0.2437
	High = 1	0.0249	0.0464	0.4618	22.0104	0.2546
	Diff	0.00***	0.00***	0.00*	-0.24***	-0.01**
	T-stat	-5.56	-8.21	-1.68	-28.04	-2.25

Table 5 reports the relationship between Centrality variables and firm performance. In the two sample T-test, we use ***, **, * represent statistical significance level at 1%, 5% and 10%, respectively. ROA refers to Return on Assets. ROE refers to Return on Equity. DEGREE is measuring the degree centrality. CLOSENESS is measuring the closeness centrality. BETWEENNESS is measuring the betweenness centrality. EIGENVECTOR is measuring the eigenvector centrality.

6 EMPIRICAL ANALYSIS

6.1 OLS regression model

To quantitively examine the relationship between constructed centrality measure and firm performance ratios, we design the empirical model as stated in equation (1):

$$Firm\ Performance = \alpha + \beta \times Centrality + Controls + \varepsilon \qquad (1)$$

The depended variables are ROA, the return on asset ratio, ROE, the return on equity ratio. . .. The focused independent variables are four centrality measure. . . .To control for industry and year difference, we add industry and year fixed effect in our model. Further, we adjust the estimated standard errors by clustering at firm level.

Table 6 reports the estimation result of equation (1). Column (1) does not include any centrality measures; column (2) to (5) include each of four centrality measures and column (6) includes all orthogonalized centrality measure. By including centrality measures into model, the adjusted R2 increases by 0.17% from 0.0262 to 0.0279, as exhibited in column (1) and (6). This suggests that our centrality measures can improve model fitness/performance by 6.48% (=0.17%/2.62%). In column (2), the controlled centrality is DEGREE_M. The estimated coefficient is 1.5662, which is statistically significant at 5% level (T-stat=2.03), indicating DEGREE_M is positively significantly related with ROA. Economically, holding other variables constant (ceteris paribus), one standard deviation increases of DEGREE_M (0.0472%) leads to 0.074% increase of ROA. Considering the mean of ROA is 2.38%, the increase of ROA is about 3.21%, which is not as trivial as it seems. In Column (3), the controlled centrality is CLOSENESS_M. The estimated coefficient is 0.0149 which is statistically significant at 1% level (T-stat=3.92), suggesting CLOSENESS_M is positively significantly related to ROA. Economically, holding other variables constant (ceteris paribus), one standard deviation increases of CLOSENESS_M (11.06%) lead to 0.16%, Considering the mean of ROA is 2.38%, the increase of ROA is 6.92%, which is as trivial as it seems. In Column (4), the controlled centrality is BETWEENNESS_M. The estimated coefficient is 18.9480, which is statistically significant at 5% level (T-stat=2.01), indicating BETWEENNESS_M is positively significantly related with ROA. Economically, holding other variables constant (ceteris paribus), one standard deviation increases of BETWEENNESS_M (0.0026%) lead to 0.049% increase of ROA. Considering the mean of ROA is 2.38%, the increase of ROA is about 0.021%, which is not as trivial as it seems.

Column (6) includes all orthogonalized centralities into regression. We find that when adding those orthogonalized centralities, BETWEENNESS_M turns insignificant anymore, yet DEGREE_M and CLOSENESS_M remain significant at 5% and 1%, respectively. This indicates major variations of BETWEENNESS_M can be explained by DEGREE_M and CLOSENESS_M.

7 ROBUSTNESS ANALYSIS

To show our results holds in various circumstances, we conduct a series of robustness checks. Our results are robust when adding more controls such as quarter fixed effect, firm fixed effect etc. If

Table 6: Empirical Estimation Result

	(1)	(2)	(3)	(4)	(5)	(6)
	ROA	ROA	ROA	ROA	ROA	ROA
DEGREE_M		1.5662**				
		(2.03)				
CLOSENESS_M			0.0149***			
			(3.92)			
BETWEENNESS_M				18.9480**		
				(2.01)		
EIGENVECTO~M					0.0229	
					(0.42)	
DEGREE_O						0.0007**
						(2.06)
CLOSENESS_O						0.0015***
						(3.48)
BETWEENNESS_O						0.0003
						(1.22)
EIGENVECTO~O						-0.0001
						(-0.42)
_CONS	0.0286***	0.0285***	0.0283***	0.0286***	0.0286***	0.0291***
	(8.94)	(8.89)	(8.90)	(8.94)	(8.93)	(9.14)
Year FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
R2	0.0262	0.0265	0.0278	0.0264	0.0262	0.0279
N	153548	153548	153548	153548	153548	153548

Table 6 reports the OLS estimation results by regression centralities on ROA. The dependent variable is ROA, which is firm return on asset ratio. In column (1), the independent is DEGREE_M, which is the firm level degree centrality. Col (1) to (4) controls industry and year fixed effects. The standard errors are estimated by clustering at firm level. All continuous variables are winsorized at 1% level. ***, **, * indicate significance level at 1%, 5% and 10%, respectively. The dataset is from 2001 to 2019.

we change of centrality measure from raw standardized measure to ranking measure following Larcker et al., (2013) to solve the issue of incomparability between centralities, our main conclusions remain the same. Results are available upon request.

8 CONCLUSIONS AND OUTLOOKS

This paper collects individual profile data and constructed social networks for China-listed firms' managers. By computing four centrality measures, Degree Centrality, Closeness Centrality, Betweenness Centrality and Eigenvector Centrality, we constructed centrality indicators for each firm. Through univariate analysis and regression empirical analysis, we demonstrate that those constructed centrality measures are significantly related with firm performance indicators.

Comparing to the existed research papers, our research is much different than others from three aspects. Firstly, the uniqueness of our data. Our data is based on Chinese company management. Secondly, diversity of measurement. We have multiple measurement the centrality, they are Closeness, Betweenness, Degree and Eigenvector. Our discussion from all four aspects makes our research more comprehensive. Thirdly, we compare management's centrality to the data of company's performance, which better reflect a managers importance related to company performance.

This paper is not flawless. The way we measure the variables is not perfect. There are some drawbacks in this research. The year we used to connect the managers are only 2016, which is the most recent available data in CSMAR dataset, instead of all years reported we had from 2001 to 2016. This may cause several potential problems for us. Firstly, we may introduce future data in analysis for sample 2001 to 2016. This problem can be overcome by introducing all historical data when constructing the social network. Nevertheless, this approach will significantly increase the computation and storage requirements. Secondly, the centrality may change over years. If we use a single year's centrality to represent the overall centrality in all time periods, it may lead to some generalizability concerns. Thirdly, we use the simple average of managers of centrality scores to represent the overall centrality of a firm. Yet, managers may have different weights inside a firm.

Nonetheless, this paper attempts to link the manager centrality and firm performance and to answer whether centrality measures are correlated with firm performance indicators. It directs future research on this stream with additional thoughts.

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